

Mark schemes

Q1.

(a) (i) $(1+x)^{-1} = 1 + (-1)x + px^2 + qx^3$
 $p \neq 0, q \neq 0$

M1

$$= 1 - x + x^2 - x^3$$

$$SC \ 1/2 \text{ for } = 1 - x + px^2$$

A1

2

(ii) $(1+3x)^{-1} = 1 - 3x + (3x)^2 - (3x)^3$

x replaced by 3x in candidate's (a)(i); condone missing brackets

M1

$$= 1 - 3x + 9x^2 - 27x^3$$

CAO SC x^3 -term: $1 - 3x + \frac{3}{9}x^2$ $\frac{1}{2}$

A1

2

Alt (starting again)

$$(1+3x)^{-1} = 1 - (3x) + \frac{(-1)(-2)(3x)^2}{2!} + \frac{(-1)(-2)(-3)(3x)^3}{3!}$$

*condone missing brackets
 accept 2 for 2!, 3.2 for 3!*

(M1)

$$= 1 - 3x + 9x^2 - 27x^3$$

CAO

(A1)

(2)

(b) $\frac{1+4x}{(1+x)(1+3x)} = \frac{A}{1+x} + \frac{B}{1+3x}$

*correct partial fractions form, and
 multiplication by denominator*

M1

$$1 + 4x = A(1 + 3x) + B(1 + x)$$

$$x = -1, \quad x = -\frac{1}{3}$$

Use (any) two values of x to find A and B

m1

$$A = \frac{3}{2}, \quad B = -\frac{1}{2}$$

A and B both correct

A1

3

Alt:

$$\frac{1+4x}{(1+x)(1+3x)} = \frac{A}{1+x} + \frac{B}{1+3x}$$

correct partial fractions form, and
multiplication by denominator

(M1)

$$1 + 4x = A(1 + 3x) + B(1 + x)$$

$$A + B = 1, \quad 3A + B = 4$$

Set up and solve

(m1)

$$A = \frac{3}{2}, \quad B = -\frac{1}{2}$$

A and B both correct

(A1)

(3)

(c) (i)
$$\frac{1+4x}{(1+x)(1+3x)} = \frac{3}{2(1+x)} - \frac{1}{2(1+3x)}$$

M1

$$= \frac{3}{2}(1 - x + x^2 - x^3) - \frac{1}{2}(1 - 3x + 9x^2 - 27x^3)$$

multiply candidate's expansions by A and
 B , and expand and simplify

m1

$$= 1 - 3x^2 + 12x^3$$

CAO

SC A and B interchanged, treat as miscopy. $(1 - 4x + 13x^2 - 40x^3)$

A1

3

Alt

$$\frac{1+4x}{(1+x)(1+3x)} = (1+4x)(1+x)^{-1}(1+3x)^{-1}$$

$$= (1+4x)(1-x+x^2-x^3)(1-3x+9x^2-27x^3)$$

write as product, using expansions condone missing brackets on $(1+4x)$ only

(M1)

$$= 1 - 4x + 13x^2 - 40x^3 + 4x - 16x^2 + 52x^3$$

attempt to multiply the three expansions up to terms in x^3

(m1)

$$= 1 - 3x^2 + 12x^3$$

CAO

(A1)

(3)

(ii) $|x| < 1$ and $|3x| < 1$

OE and nothing else incorrect

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M1

$$|x| < \frac{1}{3} \quad (0.33)$$

OE Condone $\leq$

A1

2

[12]

Q2.

(a) (i) $(1-x)^{-1} = 1 + (-1)(-x) + \frac{(-1)(-2)}{2} (-x)^2$
First two terms + kx^2

M1

$$= 1 + x + x^2$$

A1

2

$$(ii) \quad \frac{1}{(3-2x)} = \frac{1}{3} \left(1 - \frac{2}{3}x \right)^{-1}$$

Or directly substitute into formula;

B1

$$\approx * \left(1 + \frac{2}{3}x + \left(\frac{2}{3}x \right)^2 \right)$$

M1 power of 3

M1 other coefficients (allow one error)

A1 CAO

M1

$$\approx \frac{1}{3} + \frac{2}{9}x + \frac{4}{27}x^2$$

AG convincingly obtained

A1

3

$$(b) \quad (1-x)^{-2} = 1 + (-2)(-x) + \frac{(-2)(-3)(-x)^2}{2}$$

M1

$$= 1 + 2x + 3x^2$$

A1

2

$$(c) \quad 2x^2 - 3 = A(1-x)^2 + B(3-2x)(1-x) + C(3-2x)$$

M1

$$x = 1 \quad -1 = C \times 1 \quad x = \frac{3}{2} \quad \frac{3}{2} = A \times \frac{1}{4}$$

Or by equating coefficients

M1 same

A1 collect terms

M1 equate coefficients

A1 2 correct

A1 3 correct

M1

$$C = -1 \quad A = 6$$

Follow on A and C

A1

$$x = 0 \quad (-3 = 6 + 3B - 3)$$

or other value $\Rightarrow$ equation in A, B, C

m1

$$B = -2$$

A1

5

$$(d) \quad \frac{6}{3-2x} - \frac{2}{1-x} - \frac{1}{(1-x)^2}$$

$$\approx \frac{6}{3} \left(1 + \frac{2}{3}x + \frac{4}{9}x^2 \right) - 2(1 + x + x^2)$$

Follow on A B C and expansions

M1A1F

$$- (1 + 2x + 3x^2) \approx -1 - \frac{8}{3}x - \frac{37}{9}x^2$$

CAO

A1

3

[15]

Q3.

$$(a) \quad (1 - 2x)^4 = (1)^4 + 4(1)^3(-2x) + 6(1)^2(-2x)^2 + [4(1)(-2x)^3 + (-2x)^4]$$

Any valid method as far as term(s) in x and term(s) in x^2 .

M1

$$= [1] - 8x + 24x^2 + [-32x^3 + 16x^4]$$

$p = -8$. Accept $-8x$ even within a series.

A1

$q = 24$. Accept $24x^2$ even within a series.

A1

3

(b) x term is $\binom{9}{1} 2^8 x$
OE

M1

Coefficient of x term is $= 9 \times 2^8 = 2304 (=k)$

Condone 2304x

A1

2

(c) $(1 - 2x)^4(2 + x)^9 = (1 + px + \dots)(2^9 + kx \dots)$
Uses (a) and (b) oe (PI)

M1

$= \dots$

$= \dots + kx + px(2^9) + \dots$

Multiply the two expansions to get x terms

M1

Coefficient of x is $k + 512p = 2304 - 4096 = -1792$

ft on candidate's values of k and p .

Condone -1792x

SC If 0/3 award B1ft for $p + k$ evaluated

A1ft

3

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[8]

Q4.

(a) $(1 - x)^{-3} = 1 + (-3)(-x) + \frac{(-3)(-4)(-x)^2}{2}$
 $1 \pm 3x + x^2$ term

M1

$= 1 + 3x + 6x^2$

A1

2



(b) $\left(1 - \frac{5}{2}x\right)^{-3} = 1 + 3\left(\frac{5}{2}x\right) + 6\left(\frac{5}{2}x\right)^2$
 $x \rightarrow \frac{5}{2}x, \text{ incl. } \left(\frac{5}{2}x\right)^2$ *seen or implied*
(or start again)

M1

$$= 1 + \frac{15}{2}x + \frac{75}{2}x^2$$

CAO OE

A1

2

(c) $\left|\frac{5}{2}x\right| < 1 \quad |x| < \frac{2}{5}$

Sight of $\frac{\pm 5}{2} \text{ or } \frac{\pm 2}{2}$

M1A1

(d) $= 8\left(1 + \frac{15}{2}x + \frac{75}{2}x^2\right) = 8 + 60x + 300x^2$

$k \times \text{their } \left(1 - \frac{5}{2}x\right)^{-3}$

M1

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ft only on 8 $\left(1 - \frac{5}{2}x\right)^{-3}$

A1F

2

Alternative, start again:

$$8 \times \text{expression or } k \times \left(1 - 3\left(\pm \frac{5}{2}x\right)\right) \quad (\text{M1})$$

CAO (A1)

[8]

Q5.

(a) (i) $(2 + x)^3 =$

M1

$$(2^3) + 3(2^2)(x) + 3(2)(x^2) + (x^3)$$

A1

$$\dots = 8 + 12x + 6x^2 + x^3 \quad (*)$$

Any valid method; must contain all components

Accept $a = 12$

Accept $b = 6$

A1

3

$$(ii) \quad (2 - x)^3 = 8 - 12x + 6x^2 - x^3 \quad (**)$$

Clear $x \rightarrow -x$ in (i) OE

M1

ft numerical a and b

A1ft

2

$$(b) \quad (2 + x)^3 - (2 - x)^3 = (*) - (**)$$

Subtracts the 2 expressions in (a)

M1

$$\dots\dots = 24x + 2x^3$$

CSO AG (be convinced)

A1

2

$$(c) \quad \frac{dy}{dx} = 24 + 6x^2$$

A power of x decreased by 1

M1

$$\text{For st. pt. } 24 + 6x^2 = 0$$

A1

$$\text{Not possible since } 24 + 6x^2 > 0$$

Any valid explanation

E1

3

[10]

Q6.

(a) $(1 + x)^4 = 1 + 4x + 6x^2 + 4x^3 + x^4$

Full method

M1

A1 if four terms correct or just one slip

A2,1

3

(b) (i) $(1 + \sqrt{5})^4 = 1 + 4\sqrt{5} + 6(\sqrt{5})^2 + 4(\sqrt{5})^3 + (\sqrt{5})^4$
Substitute. $\sqrt{5}$ for x .

M1

$= 1 + 4\sqrt{5} + 6(5) + 4(5\sqrt{5}) + (25)$
Two of 3 terms shown in brackets

A1ft

$= 56 + 24\sqrt{5}$
AG Be convinced

A1CSO

3

(ii) $\log_2 (1 + \sqrt{5})^4 = \log_2 [8(7 + 3\sqrt{5})]$

M1

$= \log_2 8 + \log_2 (7 + 3\sqrt{5})$

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$= 3 + \log_2 (7 + 3\sqrt{5})$
SC B1 Change to base 10 and verify as per model marked script

A1CSO

3

[9]

Q7.

(a) $(1 + x)^{-\frac{1}{2}} = 1 + \left(-\frac{1}{2}\right)x + \frac{\left(-\frac{1}{2}\right)\left(-\frac{3}{2}\right)}{2}x^2$



$$1 - \frac{1}{2}x + \frac{3}{8}x^2 \quad \text{but simplification not required}$$

M1A1

2

(b) $\frac{1}{\sqrt{1+2x}} = (1+2x)^{-\frac{1}{2}}$

B1

$$= 1 - \frac{1}{2}(2x) + \frac{3}{8}(2x)^2$$

Condone missing brackets, if recovered

M1

$$= 1 - x + \frac{3}{2}x^2$$

CAO

A1

3

(c) $1 - (-0.1) + \frac{3}{2}(-0.1)^2 (= 1.115)$

Attempt to substitute in

M1

$$(1 - 0.2)^{-\frac{1}{2}} = \frac{\sqrt{5}}{2}; \quad 2 \times 1.115 = 2.23 \approx \sqrt{5}$$

Link between $\frac{1}{\sqrt{1+2x}}$ and $\sqrt{\frac{5}{2}}$

M1

AG; convincingly obtained

A1

3

[8]